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| #-- mode: org -- 96120: THE DEGREE OF THE LINEAR ORBIT OF<br>A CUBIC SURFACE (join work with SASCHA TIMME and MADELEINE<br>WEINSTEIN) |                                                                                                    |          |

# 1 Setting

## 1.1 Base field: $\mathbb{C}$

## 1.2 Cubic surfaces $S \subseteq \mathbb{P}_{x,y,z,w}^3$

$S = V_+(F) \subseteq \mathbb{P}^3$  with  $F = a_{300}x^3 + a_{210}x^2y + \dots$

There are 20 coefficients, so its parameter space is  $F = [a_{300} : \dots : a_{003}] \in \mathbb{P}^{19}$

## 1.3 Linear change of coordinates $\varphi \in \mathrm{PGL}(\mathbb{C}, 4)$ acting on $\mathbb{P}^{19}$

$\varphi : \mathbb{P}^3 \rightarrow \mathbb{P}^3$  defines an action on  $\mathbb{P}^{19}$

$$\varphi = \begin{pmatrix} A_{1,1} & A_{1,2} & A_{1,3} & A_{1,4} \\ A_{2,1} & A_{2,2} & A_{2,3} & A_{2,4} \\ A_{3,1} & A_{3,2} & A_{3,3} & A_{3,4} \\ A_{4,1} & A_{4,2} & A_{4,3} & A_{4,4} \end{pmatrix} \in \mathrm{PGL}(\mathbb{C}, 4) \subseteq \mathbb{P}^{4 \times 4 - 1}$$

$$\tilde{\varphi} : \mathbb{C}[x', y', z', w'] \rightarrow \mathbb{C}[x, y, z, w]$$

sending  $x'$  to  $A_{1,1}x + A_{1,2}y + A_{1,3}z + A_{1,4}w$  and so on.

We have  $\varphi(S) = V_+(F \circ \tilde{\varphi})$  is another cubic surface.

So,  $\mathrm{PGL}(\mathbb{C}, 4)$  acts on  $\mathbb{P}^{19}$  by precomposition

$$\varphi \cdot F := F \circ \tilde{\varphi}$$

### 1.3.1 Example 2.1

Pairs of points in  $\mathbb{P}_{x,y}^1$

$$\tilde{\varphi} : \mathbb{C}[x, y] \rightarrow \mathbb{C}[x, y]$$

$$\varphi \cdot f = \begin{pmatrix} a_{11}^2 & a_{11}a_{21} & a_{21}^2 \\ 2a_{11}a_{12} & a_{11}a_{22} + a_{12}a_{21} & 2a_{21}a_{22} \\ a_{12}^2 & a_{12}a_{22} & a_{22}^2 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

### 1.3.2 More compactly

We consider a  $\mathbb{C}$  vector space  $V$  of dimension 4. Given  $\varphi \in \mathrm{PGL}(V)$  and  $F \in \mathbb{P}\mathrm{Sym}^3(V)$ ,

$$\varphi \cdot F := \mathrm{Sym}^3(\varphi)(F)$$

## 1.4 The variety $\Omega_F$ , closure of $\mathrm{PGL}(\mathbb{C}, 4) \cdot F$

We do *not* know its equations, but we have the morphism

$$\Theta_F : \mathrm{PGL}(\mathbb{C}, 4) \rightarrow \Omega_F \subseteq \mathbb{P}^{19}$$

sending  $\varphi$  to  $\varphi \cdot F$

- It is *dominant* (trivial)
- It is one-to-one (a general  $F$  has trivial stabilizer)
  - Let me call it “the parametrization”
  - $\dim(\Omega_F) = \dim(\mathrm{PGL}(\mathbb{C}, 4)) = 15$

## 1.5 Goal: Compute the degree of $\Omega_F$

Motivation: revisit cubic surfaces with modern techniques.

- challenge modern techniques
- what can be said about

Question 1 of

TWENTY-SEVEN QUESTIONS ABOUT THE CUBIC SURFACE (KRISTIAN RANESTAD - BERND STURMFELS)

# 2 Formulations

## 2.1 Common part

### 2.1.1 Packages

---

```
1 using HomotopyContinuation
2 using LinearAlgebra, PolynomialRoots, Combinatorics
```

---

### 2.1.2 Coordinates of $\mathbb{P}^3$ and $\mathrm{PGL}(\mathbb{C}, 4)$

---

```
1 @var x y z w
2 @var A[1:4, 1:4]
```

---

### 2.1.3 General cubic surface

---

```
1 monomials = [prod([x,y,z,w].^exp) for exp in multiexponents(4,  
    ↪ 3)]  
2 c = randn(ComplexF64, 20);  
3 F = sum(c .* monomials);
```

---

### 2.1.4 The “parametrization” $\Theta_F : \mathrm{PGL}(\mathbb{C}, 4) \rightarrow \Omega_F$

---

```
1 = coefficients(subs(F, [x,y,z,w] => A*[x,y,z,w]), [x,y,z,w])  
2 typeof()  
3 size()
```

---

## 2.2 Formulation 1: 15 general hyperplanes

### 2.2.1 General linear space in $\mathbb{P}^{19}$ of dim 4

---

```
1 L = randn(ComplexF64, 15, 20);
```

---

### 2.2.2 Pulling back to $\mathrm{PGL}(\mathbb{C}, 4)$

---

```
1 Eq1 = L*;  
2 size(Eq1)
```

---

### 2.2.3 Solving

---

```
1 System1 = System(Eq1; variables=vec(A));
```

---

Bezout bound

---

```
1 solve(System1, start_system=:total_degree)
```

---

Tracking 14348907 paths... 0% | ETA: 29.85 days  
BKK bound

---

```
1 solve(System1,start_system=:polyhedral)
```

---

ERROR: InterruptException:

#### 2.2.4 Pros and Cons

- (con) In fact, the build  $\Theta$  has source  $\mathbb{P}^{15}$ . It is a rational morphism (so, System1 contains its base locus:  $\det(A) = 0$ )
  - Classify solutions. Problem: pick tolerance
  - Use monodromy method. Problem: which parameters
- (pro) Better suited for the trace test (we will see it later).

### 2.3 Formulation 2: 15 general points (monodromy)

#### 2.3.1 Idea

Given a point  $p = [x_0 : y_0 : z_0 : w_0] \in \mathbb{P}^3$ , the set of cubics passing through  $p$  form a hyperplane in  $\mathbb{P}^{19}$

$$\Pi_p = V_+(x_0^3 a_{3000} + x_0^2 y_0 a_{2100} + \cdots + w_0^3 a_{0003}) \subseteq \mathbb{P}^{19}$$

Defines a morphism

$$\mathbb{P}^3 \rightarrow \check{\mathbb{P}}^{19}$$

(the Veronese map of deg 3)

**Caution** KleimanBertini theorem ensures these hyperplane are general enough! (for a general  $p \in \mathbb{P}^3$ ,  $\Pi_p$  and  $\Omega_F$  intersect transversally)

#### 2.3.2 Coordinates for 15 points

---

```
1 @var P[1:15,1:4]
2 # parameters to do monodromy loops
3 p = P[:,4]
```

---

### 2.3.3 Equations with parameters

---

```
1 Eq2 = [subs(F, [x,y,z,w] => A*P[i,:]) for i in 1:15];
```

---

### 2.3.4 Computing initial pair solution-parameter

1. Fix random parameters

We pick 15 random points (and leave one coordinate per point unspecified)

---

```
1 P = randn(ComplexF64, 15, 3);
```

---

2. System of eq. for monodromy

---

```
1 Eq2_mono = subs(Eq2, A[4,4] => 1, vec(P[:,1:3]) =>
  ↪ vec(P));
```

---

3. Initial solution: random linear transformation.

---

```
1 A = randn(ComplexF64, 4, 4)
2 A[4,4] = 1
3 # Initial solution for the affine equations.
4 a = vec(A)[1:15]
```

---

4. Compute initial parameters: the last coordinate

---

```
1 p = map(Eq2_mono) do f
2   # substitute and compute root
3   subs_f = subs(f, vec(A) => vec(A))
4   last(roots(coefficients(subs_f, variables(subs_f)) |>
    ↪ reverse))
5 end; # initial value of the parameters
```

---

### 2.3.5 Solving monodromy

---

```
1 System2 = System(Eq2_mono, variables=vec(A)[1:15],  
  ↪ parameters=p)  
2 res = monodromy_solve(System2, a, p)
```

---

```
## Solutions found: 96120 Time: 0:45:35 ## MonodromyResult ##  
===== ## 96120 solutions (0 real) ##  
return code heuristicstop ## 1263490 tracked paths ## seed 186832
```

## 3 Now what?

Can we do something more than simply have faith in the computations?

Yes, we can! (what a time to be alive...)

### 3.1 Alpha certificate (lower bound)

**Approximate zero** Given a polynomial system  $F = 0$ , a point  $p \in \mathbb{C}^n$  is an *approximate zero* of  $F = 0$  if the Newton method starting at  $p$  converges quadratically to a (unique) zero of  $F = 0$ .

Smale's  $\alpha$  theorem:

- Our case: We just want to ensure that our points actually approximate distinct zeros!

### 3.2 Trace test (upper bound)

Trace test

## 4 Classical proof

Previous work on plane curves LINEAR ORBITS OF SMOOTH PLANE CURVES

Using the same techniques Towards the degree of the  $\mathrm{PGL}(4)$ -orbit of a cubic surface

Classical proof with modern techniques A UNIVERSAL FORMULA FOR COUNTING CUBIC SURFACES

## 4.1 Misc

Lorde Jorja Smith Erika de Casier Japanese breakfast Ela Minus